SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Nabil The Goat
Project / Solution Name	sabave.ai
Name (Leader)	Nabil Hilmi Tsaqifi
Email & Phone (Leader)	nabiltsaqifi@gmail.com / +60 11 7244 6398
University Affiliation(s)	Monash University Malaysia
Live Prototype URL	https://sabave.web.app
Code Repository URL (public / access granted to judges)	https://github.com/Joebilly112/sabaveai

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

User (Owner or Staff, in the browser) → React 19 + Vite web app on Firebase Hosting (sabave.web.app) → /api/** rewrite → Express 5 API running as a Firebase Cloud Functions (2nd gen, Node 22) function.

Inside the API: (1) the weather signal engine calls the Open-Meteo Air Quality API (Copernicus CAMS PM2.5) for each outlet's coordinates, with an offline forecast fallback; (2) the trigger engine opens an alert when a forecast day reaches Unhealthy; (3) deterministic campaign rules read the business's Brand Kit and pick options and budgets; (4) the copy generator calls the DeepSeek chat-completions API (deepseek-v4-pro) server-side for Bahasa Malaysia and English captions, validated against a never-say list, with template fallback; (5) the approval service enforces Owner-only approval and budget caps; (6) a Meta connector publishes (simulated); (7) every step is written to the activity log.

Data (businesses, alerts, drafts, log) is held in an in-memory store in the prototype; a Drizzle/PostgreSQL schema is scaffolded for persistence. API keys are server-side secrets and never reach the browser.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9	Main language for frontend and backend	Apache-2.0
React 19.1 + React DOM	Web app user interface	MIT
Vite 7.3	Frontend build tool and dev server	MIT
Tailwind CSS 4.3 + shadcn/ui (Radix UI)	Styling and UI components	MIT
Express 5.3 (Node.js 22)	Backend REST API	MIT

Name & Version	Purpose in Project	License
TanStack React Query 5, wouter 3, Zod 3, Orval 8	Data fetching, routing, validation, typed API client	MIT; Unlicense (wouter)

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Open-Meteo Air Quality API (Open-Meteo / Copernicus CAMS)	Hourly PM2.5 forecast per outlet; triggers haze alerts	Free
DeepSeek Chat Completions API (DeepSeek)	Writes Bahasa Malaysia and English captions; brand-style summaries	Paid
Meta Graph API (Meta), simulated connector	Publishing to Facebook/Instagram; simulated in prototype	Not yet connected
Leaflet map tiles (OpenStreetMap-based)	Dashboard map of the outlet location	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Firebase Hosting (Google)	Serves the web app at sabave.web.app	Blaze (pay-as-you-go)
Firebase Cloud Functions 2nd gen / Cloud Run (Google)	Runs the Express API (function “api”, Node 22)	Blaze / us-central1
Google Cloud secret environment variables	Stores the DeepSeek API key server-side	Blaze / us-central1
GitHub	Source code repository and version history	Free
Replit (Phase 1 only)	Development and hosting of the preliminary prototype	Replit Core (sponsor)

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek V4 Pro (deepseek-v4-pro)	DeepSeek, via API	Pre-trained	DeepSeek API Terms of Service

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Open-Meteo / Copernicus CAMS PM2.5 forecast	https://open-meteo.com	Public	CC BY 4.0	N
Offline haze forecast (fallback)	data/offlineForecast.json in repo	Synthetic	Team-created	N
Seed businesses (kopitiam, laundromat, boutique)	data/businesses.json in repo	Synthetic	Team-created	N
DOSM Economic Census 2023 and MSME 2024 statistics	https://www.dosm.gov.my	Public	DOSM open data terms	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude (Anthropic), Opus model	Research, pitch deck, documentation, reviewing ideas, writing prompts for the coding agents	Major
Replit Agent (Replit)	Generating code for the Phase 1 prototype from the founder's specifications	Major
Cursor (Agent mode)	Phase 2 code changes: pages, onboarding, demo accounts, Firebase deployment	Major
DeepSeek (in-app)	Writes captions inside the product (declared in 4.1), not used to build	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The founder reviewed every AI-generated change before committing it. The core rules (triggers, campaign options, budgets, approval) are deterministic and covered by four Node test-runner suites (alert thresholds, non-delivery options, never-say blocking, Owner-only approval, AI fallback, input validation), which were run after changes. The full demo journey was tested end to end for each business type. Research figures were checked against their original sources and cited. The founder is responsible for all design decisions, logic and claims in the submission.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform (Replit Core)	Built, stored secrets for and hosted the Phase 1 prototype; replaced by Cursor and Firebase for Phase 2 after Replit became unavailable

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Not essential. Replit was used to develop and host the Phase 1 prototype. When Replit became unavailable for the finals, the same codebase was moved to Cursor and Firebase without functional changes, which shows the prototype does not depend on the sponsor resource.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No customer personal data is collected. The app stores business-level information entered by the owner (business name, city and coordinates, products and prices, menu photos, delivery options, channels, budget cap, never-say list) and staff display names and roles. Only business data (products, prices, tone, offer) is sent to the DeepSeek API to write captions; brand analysis sends text only. Outlet coordinates are sent to Open-Meteo for the forecast.

In the prototype, data lives in server memory and is cleared when the server restarts; the demo uses synthetic businesses. API keys are server-side secrets, never in the repository or the browser. In production we would add consent at onboarding, authenticated role-based access, encrypted managed storage and a fixed retention period, in line with PDPA 2010 principles.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

AI copy can be wrong or off-brand. Mitigation: the model writes captions only, never offers, budgets or publish decisions; its JSON is validated, checked against the never-say list, retried once and replaced by template copy if it fails, and every draft needs Owner approval. Dependency on one LLM provider: the client is OpenAI-compatible and configured by environment variables, so the provider can be swapped. Forecasts can be wrong: the owner decides, and an offline forecast is used if the API fails. Other limits: in-memory storage resets on restart, publishing to Meta is simulated until API approval, and roles use local accounts rather than full authentication.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



11/1/2026